

R- PROGRAMMING CODE OUTPUTS-7,8,9

7)OUTPUT-

RStudio interface showing the source editor with R code and the Environment pane displaying the results of an ANOVA test.

```
> df <- read.csv(file.choose())
>
> df$ethnicity <- as.factor(df$ethnicity)
> anova_result <- aov(age ~ ethnicity, data = df)
> summary(anova_result)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ethnicity	4	598019	149505	403.8	<2e-16 ***
Residuals	23700	8775524	370		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The Environment pane shows the following data objects:

Object	Description
age_gender	23705 obs. of 5 variables
anova_result	Large aov (13 elements, 1.9 MB)
df	23705 obs. of 5 variables

8)OUTPUT-

RStudio interface showing the source editor with R code and the Environment pane displaying the results of an ANOVA test.

```
> df <- read.csv(file.choose())
>
> df$Gender <- as.factor(df$Gender)
> df$Study_Method <- as.factor(df$Study_Method)
> anova_result <- aov(Score ~ Gender * Study_Method, data = df)
> summary(anova_result)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Gender	1	102.08	102.08	31.410	0.00138 **
Study_Method	2	153.50	76.75	23.615	0.00143 **
Gender:Study_Method	2	3.17	1.58	0.487	0.63671
Residuals	6	19.50	3.25		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The Environment pane shows the following data objects:

Object	Description
age_gender	23705 obs. of 5 variables
anova_result	List of 13
bitcoin_price_lwe...	7 obs. of 7 variables
df	12 obs. of 3 variables
student_scores_tw...	12 obs. of 3 variables

R- PROGRAMMING CODE OUTPUTS-7,8,9

9)OUTPUT-

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for reading a CSV file, creating a contingency table, and performing a chi-squared test. The output shows the test results.
- Environment Pane:** Lists objects in the Global Environment, including 'age_gender', 'anova_result', 'bitcoin_price_1we...', 'chisq_result', 'course_preference', 'df', and 'student_scores_tw...'. Each entry shows its data type and dimensions.
- Files Pane:** Shows the file structure, including a folder named 'Module_2'.
- Taskbar:** Displays the Windows taskbar with various application icons and system information (29°C Sunny, 12:01, 05-01-2026).

```
R - R 4.5.2 ~ /
> df <- read.csv(file.choose())
>
> table_data <- xtabs(Count ~ Gender + course, data = df)
>
> chisq_result <- chisq.test(table_data)
>
> chisq_result

Pearson's Chi-squared test

data:  table_data
X-squared = 8.8095, df = 2, p-value = 0.01222
> |
```

Object	Value
age_gender	23/05 obs. of 5 variables
anova_result	List of 13
bitcoin_price_1we...	7 obs. of 7 variables
chisq_result	List of 9
course_preference	6 obs. of 3 variables
df	6 obs. of 3 variables
student_scores_tw...	12 obs. of 3 variables